GEB: A Computable System for Group Organization

Abstract

Current AI and blockchain systems are generally facing a computability category crisis because they are rooted in the paradigm of a single formal system represented by the "individual Turing machine." This category misplacement leads to a logical dilemma: attempting to drive an evolutionary system that essentially requires **completeness** by relying on algorithms that pursue absolute **consistency**. This mismatch makes spontaneous system convergence difficult and eventually forces the introduction of a super-observer to impose order.

This paper argues that the theoretical root of this dilemma lies in **Godelian incompleteness**. To break through this limit, we integrate the intellectual trajectories of Godel, Turing, Feferman, and Nash, and propose a computability theory for group organization: the GEB framework. This theory constructs a spacetime-orthogonal evolutionary model: in the **horizontal spatial dimension**, non-cooperative games among individuals drive competition and determine the system's state distribution at any instant; in the **vertical temporal dimension**, recursive superposition of system states collapses instantaneous probabilistic fluctuations into deterministic physical facts.

The model is ultimately transformed into a four-element collaborative architecture consisting of individual accounts, Turing machines, non-cooperative games, and ordinal logic. Specifically: **individual accounts** establish sovereignty boundaries; **Turing machines** define the boundaries of physical laws; **non-cooperative games**, as spatial convergence operators, drive the system to search for local optima through individual competition; **ordinal logic**, as a temporal evolution operator, provides irreversible historical depth through state recursion.

Ultimately, GEB theory not only provides a unified perspective for understanding the essence of Bitcoin, but also offers profound theoretical insights and an engineering blueprint for building **adaptive computable systems of group organization**.

1. Introduction: The Category Crisis of Computability

In today's rapid development of AI and blockchain, although both technologies have made significant progress in their respective fields, their underlying architectures remain constrained by a single formal-system paradigm. When facing the complexity of the real world, this paradigm exposes deep adaptive deficiencies. We argue that this deficiency does not stem from performance or algorithmic shortcomings, but from a foundational **category crisis of computability**.

1.1 Current Landscape: Structural Dilemmas of a Single Formal System

Mainstream AI and blockchain systems are, in essence, built on a closed computational paradigm grounded in the abstraction of the "individual Turing machine." This paradigm pursues global consistency of symbolic deduction, yet shows obvious structural limits when handling complex social relations and dynamic environments.

Limitations of AI Systems

Systems represented by large language models (LLMs) are built on symbolic mapping structures over static corpora.

- **Lack of environmental coupling:** The system is closed within the semantic space defined by training data and finds it difficult to adjust its own evolutionary trajectory in response to real-time external changes.
- **Lack of autonomous verification:** The model cannot endogenously determine the truthfulness of its outputs; at its core, it remains a probabilistic generation mechanism in a closed symbolic system rather than a cognitive system anchored to reality.
- **Centralized evolutionary path:** System optimization depends on external centralized compute and retraining processes, rather than continuous spontaneous evolution from within the structure.

Limitations of Blockchain Systems

Although blockchain achieves data immutability, when supporting complex behavior it shows clear structural closure and gradually depends on centralized proxy mechanisms.

- **Closed state updates:** The system cannot natively perceive off-chain information, resulting in dependence on external input mechanisms such as oracles, which themselves often introduce new trust centers.
- **Account-model constraints:** Taking Ethereum as an example, its global state structure makes individual assets depend on a unified global interpretation of state, weakening structural independence. In the end, user control over assets depends on trust in global rule execution.

1.2 Theoretical Root: Cognitive Boundaries of the Individual Turing Machine

The theoretical root of the above dilemma is that modern computational systems generally adopt the Turing machine model proposed by Alan Turing in 1936. The Turing machine abstracts computation as individual, localized symbolic operations. This laid the foundation of modern computing, but also set its insurmountable boundaries:

1. **The incompleteness dilemma:** Godel's incompleteness theorem reveals that any sufficiently complex formal system contains propositions it cannot prove within itself. This means a system operating solely on its own deterministic rules cannot understand "reality" beyond its symbolic framework.
2. **Category misplacement:** Over the past century, we have tacitly assumed that the **organizational properties of groups** can be approximated by simply aggregating **individual computational capacities**. But this assumption does not hold in complex systems.

The reason is that the key characteristics of group systems (such as path competition, consensus convergence, and historical dependence) are not functions of individual states, but products of interactions among multiple individuals. Such processes cannot be equivalently reduced within a single computational subject.

This **category misplacement** leads to a core paradox: we attempt to use deterministic algorithms that pursue **consistency** to build an evolutionary system that requires **completeness**. The result is inevitably centralized: to achieve determinacy, one must introduce a **super-observer**.

1.3 Bitcoin's Insight: Decoupling Blockchain from Bitcoin

By examining Bitcoin's evolution, we can observe a long-overlooked structural fact: **its core value does not come entirely from blockchain technology itself.** Here, "decoupling" does not mean technical removal, but conceptual distinction between two different mechanisms:

- **Blockchain:** Belongs to the category of deterministic rule systems; its core objective is consistency of symbolic verification. In essence, it is a static mechanism for state recording and validation.
- **The non-blockchain part of Bitcoin:** Manifests as a group game-based, cross-individual evolutionary process, including hashrate competition, path selection, and consensus formation.

Market feedback shows that projects pursuing blockchain technology alone often drift toward centralized consistency and lose Bitcoin's robustness. This implies that we have long interpreted a fundamentally "uncertainty-evolving" organism (Bitcoin) through a "deterministic" mindset (Blockchain).

This very misalignment reveals the shared underlying dilemma in AI and blockchain today: it is not an implementation defect of a single technology, but an inevitable structural boundary of the single-formal-system paradigm.

If we continue using individual Turing machine theory to build social organization, the inevitable outcome is centralized structures that lose evolutionary vitality. To break this limitation, we must redefine "how systems perceive reality," return to the logical source, and explore a group-oriented, dynamic, uncertainty-embracing framework for organizational computation.

Therefore, what stands before humanity is not how to make the "chain" faster, but how to awaken from the "dream of determinism" and seek a truly **computable theory of group organization** capable of supporting the evolution of productive relations.

2. GEB Theory: A Computability Framework for Group Organization

This chapter aims to construct a computability framework for group organization capable of addressing Π_2-level challenges. The theory integrates Godel's disclosure of incompleteness, Turing's extensible design of ordinal logic, Feferman's characterization of formal-system boundaries, and Nash's proof of non-cooperative game convergence, ultimately forming a structurally expressed **spacetime orthogonal duality**.

2.1 From Logical Incompleteness to Evolutionary Completeness: The Deep Challenge of Π_2 Propositions

2.1.1 Logical Blind Spots and the Π_2 Boundary

The dilemma of any complex system originates from the cognitive boundary of a single formal system. Godel proved that every self-consistent system contains truths it cannot prove from within. Feferman's work further characterizes the capability boundary of formal systems in ongoing expansion, with one typical form represented as a Π_2 proposition in standard form:

$$(\forall x)(\exists y)\ R(x, y)$$

The key characteristic of this class is: for a given input x, the system can verify whether some result y satisfies relation $R(x, y)$, but cannot pre-construct a unified method that generates all corresponding y.

In decentralized systems, a more fitting analogy is the **blockchain fork-selection problem**: for all nodes ($\forall x$), there exists a valid chain state ($\exists y$) that can be verified, yet the system cannot determine in advance which fork will be globally accepted as the unique history. This structural isomorphism is precisely the concrete projection of Π_2 propositions in decentralized environments.

Therefore, the system can **verify** truth (the P category), but cannot proactively **generate** unique historical truth (the NP category).

2.1.2 Temporalization Through Turing's Ordinal Logic

To let systems continue moving forward at undecidable points, Turing proposed **ordinal logics** in 1939. It provides an extension mechanism for formal systems: when a proposition is undecidable under the current axiomatic system, a new axiom can be introduced to move the system into a higher-level logical structure.

This process can be represented as:

$$L_{\alpha+1} = L_\alpha \cup \{A_\alpha\}$$

where A_α denotes the axiom introduced at level α. The significance of this mechanism is that computation is no longer a one-shot function solving process, but an ongoing expansion of expressive capacity along the time dimension.

2.1.3 From Logical Extension to Organizational Emergence

However, ordinal logic only addresses the formal path of "how logic evolves to higher levels," and does not answer: **in a decentralized environment, who decides which axiom (or which historical path) is introduced into the system?**

When a system faces the uncertainty generated by Π_2 propositions, pure logical deduction falls into endless path branching. For the system to collapse from "logically infinite possibilities" into "a unique history in reality," variables outside logic must be introduced, namely game competition in the spatial dimension. This means computation must evolve from pure symbolic deduction into a spacetime game anchored to physical reality.

2.2 Computability of Group Organization: A Spacetime-Orthogonal Evolutionary Model

Following orthogonal structures in physics (here "physics" is metaphorical, referring to **formal constraint rules** internal to the system rather than natural laws), GEB decomposes the above game process into two independent dimensions: the horizontal space constrains individual conflict and randomness, while vertical time solidifies structural evolution and determinacy. The core idea of this orthogonal design is: rather than trying to eliminate uncertainty on a single logical layer, it leverages uncertainty through dimensional separation, thereby collapsing deterministic historical facts from a probabilistic game substrate.

2.2.1 Horizontal Structural Dimension (Space / Non-Cooperative Game)

This dimension describes the state distribution of the system on any instantaneous cross-section. It is both the source of system **uncertainty** and the source of **structural constraints**.

- **Driving force: Inconsistency among individuals**. The system does not attempt to erase these differences; instead, multiple agents engage in non-cooperative games under local rules.
- **Mechanism:** Each participant makes probabilistic decisions based on self-interest. This competitive pressure forces the system to seek local optima at each moment rather than rely on preset rigid instructions.
- **Function:** Provides **structural constraints**. Nash equilibrium guarantees that in the presence of conflicting interests, the system does not diverge into disorder, but maintains a dynamically stable game cross-section as a logical starting point for subsequent evolution.

- **Property: Non-locality**. Organizational properties of the system (such as security and consensus) do not belong to any isolated individual, but emerge spontaneously from group interactions in the spatial dimension.

2.2.2 Vertical Evolutionary Dimension (Time / Recursive Update)

This dimension describes recursive superposition of system states over time. It is the **collapse operator** that converts "logical possibilities" into "physical facts."

- **Driving force: Recursive superposition** of states. This is not simple data accumulation, but a self-supporting structure in which later terms reference earlier terms. This recursiveness ensures each new layer is generated on the logical base of the previous one.
- **Function:** Achieves **logical collapse**. Instantaneous fluctuations generated by spatial games (such as fork paths) rapidly lose survival space in the vertical dimension as recursion depth increases. Through historical depth stacking, originally divergent multi-path uncertainty is gradually solidified into a unique deterministic path.
- **Property: Logical inertia and irreversibility**. Recursive accumulation generates a physical arrow of time. The deeper the history, the higher the "negative entropy" cost required to alter it, enabling the system to evolve from a fragile symbolic set into a self-organizing system with historical causal depth.

2.2.3 Coupling Mechanism: Probabilistic Convergence and Property Transformation

Through probabilistic mechanisms, spacetime orthogonal logic realizes transformation from "individual uncertainty" to "group certainty."

- **Probability as a coupling interface:** Spatial games output a "distribution of possibilities" (e.g., multiple competing blocks), while temporal recursion provides "path-selection pressure." Probability is not system noise, but a metric tool for **exploration and convergence** across different evolutionary paths.
- **Convergence dynamics:** The computability of group organization appears as a **probability decay process**: as vertical recursion depth d increases, conflict probability P among multiple paths generated by spatial games decays exponentially ($P \to 0$).

- **Property transformation:** This orthogonal coupling ultimately achieves a fundamental transformation of system properties: **it successfully converts localized, individual random decisions into non-localized, group-level necessary facts.**

2.3 The Four Elements of GEB: Logical Abstraction of an Evolutionary Structure

Based on spacetime orthogonal theory, GEB translates abstract logic into four complementary engineering elements. These are no longer isolated functional modules, but mutually conditioning components that jointly realize a complete closed loop of "logical collapse":

- **Individual account (localized basic particle):** Establishes a verifiable logical starting point for the system. As a spatial boundary condition, it ensures structural independence of ownership (such as UTXO) and defines the **subjects** of the game.
- **Turing machine (deterministic physical law):** Defines the computable boundary of behavior. It guarantees local consistency of "right and wrong," providing underlying **formal constraints** for upper-layer games.
- **Non-cooperative game (horizontal convergence operator):** Uses group competition as **collapse driving force**. It ensures that without a central referee, the system can leverage uncertainty to drive consensus and force multi-solution Π_2 paths to converge to unique history.
- **Ordinal logic (vertical time driver):** Builds the arrow of time through chain-recursive structure. It solidifies instantaneous game equilibria into irreversible **historical depth**, giving directional evolution to the system.

The integration of these four elements constitutes a paradigm shift in "how truth is produced": individual accounts and Turing machines maintain consistency in the "symbolic world" (P category); non-cooperative games and ordinal logic drive completeness in "real-world evolution" (NP category).

This architecture enables GEB to transcend traditional distributed ledgers, becoming a **group-organization generator** capable of self-organized evolution amid uncertainty and continuous ascent along ordinal chains. At this point, the "cognitive blind spot" introduced by Godelian incompleteness is formally transformed under a spacetime-orthogonal engineering framework into the **evolutionary driving force** of continuously ascending complex adaptive systems.

3. A Paradigm Leap: From BEVM(λ) to GEB

Chapter 2 established the theoretical framework of "spacetime orthogonal duality" from roots in mathematical logic. This chapter shifts to engineering evolution and explores how GEB, through deconstruction and recomposition of its predecessor BEVM(λ), achieves a paradigm shift from "static formal verification" to "dynamic evolutionary generation."

3.1 BEVM(λ): An Insightful Predecessor Framework

Before entering GEB, we must understand its predecessor: **BEVM(λ)**. This was our early formal attempt to translate Bitcoin from a mere "ledger engineering" system into a "pure logical language." It abstracts four core components:

- **λ calculus:** As the modeling language for state transitions and rule expression;
- **Individual model:** To characterize one-to-one mapping between user intent and state ownership;
- **Consensus algorithm:** To guarantee state validation and network-wide consistency;
- **Consensus perception mechanism:** An attempt to map off-chain energy input into on-chain value inscription.

This framework had far-reaching conceptual significance, but structurally remained a closed system. The components lacked an evolutionary coupling mechanism and could not form a complete closed loop from "deterministic rules" to "nondeterministic adjudication." In other words, BEVM(λ) resembled an excellent auditor that can precisely verify the compliance of each transaction (i.e., determine "right or wrong"), but when facing concurrent conflicts or path forks, it cannot automatically "collapse" into a unique history through internal games like a living organism.

The table below summarizes the limitations of BEVM(λ) components and GEB's corresponding improvements:

BEVM(λ) Component	Structural Limitation	Corresponding GEB Component	Core Improvement
λ calculus	Pure symbolic deduction, lacking	Turing machine	Embeds a deterministic "physical law" engine

BEVM(λ) Component	Structural Limitation	Corresponding GEB Component	Core Improvement
	formal constraint boundaries		
Individual model	Static identity description, lacking economic sovereignty	Individual account	Binds Bitcoin UTXO with sovereign independence
Consensus algorithm	Pure logical verification, no evolutionary convergence driving force	Non-cooperative game	Introduces energy/reputation-driven game convergence
Consensus perception mechanism	Unidirectional mapping, lacking temporal depth	Ordinal logic	Bidirectional recursion, constructing an irreversible arrow of time

GEB takes this as a starting point and, by redefining logical relations and evolutionary channels among components, completes the paradigm leap from a static formal system to a dynamic evolutionary system.

3.2 Engineering Reconstruction: Turning the System from "Static" to "Evolutionary"

GEB is not a simple renaming of BEVM(λ). Through the following core reconstructions, it transforms a static blueprint into a dynamically driven engine:

3.2.1 Endowing Components with "Physical Activity"

At the BEVM(λ) stage, accounts and rules were merely descriptions in code. In GEB, we transform them into internal "physical laws" that cannot be violated.

- **Action:** Deepen the Individual model into discrete, independent **individual accounts** (logical bricks), and concretize abstract modeling language into an embedded **Turing machine engine**.
- **Change:** This ensures the starting point of the game is decentralized, and all participants act under the same digital gravity (Turing-machine constraints). It provides a deterministic and solid physical foundation for upper-layer complex games.

3.2.2 Injecting "Energy-Driven Convergence"

This is the core of reconstruction, addressing the disconnection between rules and energy in BEVM(λ). We reorganize the originally loose consensus components into a coupled "time + energy" system:

- **Action:** Introduce non-cooperative game (PoW) as the system's power source. When the system faces path-selection disagreement (Π_2 proposition), it no longer relies solely on logical deduction but introduces physical energy consumption as a "clutch" to break deadlocks.
- **Change:** Participants must invest real-world computing-cost inputs for profit. This physically costly mechanism forcibly compresses and collapses logically divergent possibilities into a uniquely determined historical trunk.

Through reconstruction of BEVM(λ), GEB proves that logic is no longer symbolic relations on paper, but an entity in which **verification-computation-time-game** four-layer structures are interlocked and self-operating.

This leap enables the system to cross the boundary of mere verification and enter a new stage of "evolutionary completeness." GEB is not merely recording data; under constraints of physical law, it autonomously generates a unique real history from infinite possibilities.

4. GEB Engineering Architecture: A Heterogeneous Collaborative Agere Network

After establishing the theoretical path from "static verification" to "dynamic evolution" in the first three chapters, this chapter details GEB's concrete engineering implementation. The architecture is a collaborative network composed of multiple heterogeneous **Agere subsystems** ("Agere" comes from Latin, meaning "to act, to drive"), which precisely maps the "four elements" theory in Chapter 2 into runnable engineering components.

The overall architecture reconstructs and surpasses traditional consensus design, forming a complete closed loop from "individual verification" to "global historical convergence."

4.1 Individual Accounts: A Sovereign Foundation Symbiotic with the Bitcoin Network

The cornerstone of GEB's four elements is the "individual account." In engineering practice, GEB chooses native symbiosis with the most secure decentralized

existing system: the **Bitcoin network**.

- **Architectural implementation (native address binding):** GEB does not establish an isolated account system; its addresses are directly associated with Bitcoin addresses. User asset ownership and instruction validity are ultimately proven by the Bitcoin private keys they control.
- **Workflow (entry protocol):** By constructing a Bitcoin transaction containing specific `OP_RETURN` data, users "project" UTXO value into the GEB network. This action activates participation rights in Agere subsystems while losslessly anchoring Bitcoin security inside GEB.
- **Engineering significance:** This design realizes the "structural independence" pursued in Chapter 1. It allows GEB to natively inherit Bitcoin's decentralization, providing a unified and trustless identity foundation for upper-layer complex AI tasks.

4.2 Turing Machine: A Deterministic Formal Validation Engine

The "Turing machine" element in GEB defines the system's "physical laws." In engineering terms, it is implemented as the **deterministic verification engine (VM)** inside each Agere subsystem.

- **Architectural implementation (Agere virtual machine/rule set):** Each subsystem embeds an unambiguous instruction set according to its task type (such as AI inference verification or financial settlement logic). Its responsibility is to solve P-category problems, i.e., static auditing of task results.
- **Workflow (legality validation):** When a task proof is broadcast on the network, all nodes independently run the engine. It does not produce answers; it only checks "right or wrong" based on predefined rules.
- **Engineering significance:** It guarantees absolute internal logical consistency, ensuring determinacy of local computation and defining a bottom-layer "legal boundary" for upper-layer uncertain games.

4.3 Ordinal Logic: A Task-Driven Temporal Recursive Structure

Turing machines can judge "right/wrong" but not "before/after." The role of ordinal logic is to introduce an arrow of time into the system, expressed in engineering as a

task-driven Agere state chain.

- **Architectural implementation (task network and state chain):**
  - **Task execution:** Agere nodes listen for demands, execute Π_2-level complex tasks (such as generating code or model inference), and package results as candidate solutions.
  - **State recursion:** Once a candidate solution is confirmed as winning, it links to the previous state through hash references.
- **Workflow (witness of history):** This is precisely the instantiation of Turing ordinal logic $L_{\alpha+1} = L_\alpha \cup \{A_\alpha\}$. Every "winning solution" written into the chain obtains an irreversible temporal coordinate.
- **Engineering significance:** It organizes discrete, competitive computational tasks into a linear time sequence, preventing the possibility of multiple histories for the same task and constructing unique historical depth.

4.4 Non-Cooperative Game: A Reputation-Based Convergence Operator

This is the soul of GEB architecture, responsible for collapsing "multiple possible solutions" into "one unique truth." In engineering, this is realized through an evolutionary consensus mechanism driven by the **global reputation layer (Agere0)**.

- **Architectural implementation (Agere0 global reputation layer):**

  GEB contains a special root-level subsystem, **Agere0**. It records and maintains the global reputation values of all nodes. Reputation in GEB plays a role similar to "hashrate energy" in Bitcoin, representing long-term "negative-entropy" accumulation of node contributions to the system.
- **Workflow (convergence to unique history):**
  1. **Competition (game):** Multiple nodes submit different candidate solutions for the same Π_2 task, forming path forks.
  2. **Adjudication (collapse):** The validator network (composed of high-reputation nodes) scores and votes according to task quality.
  3. **Incentives (equilibrium):** Winners receive reputation gains, while nodes providing incorrect or low-quality solutions face reputation penalties.

- **Engineering significance:** This "cannot afford to lose" game mechanism leverages human self-interest and reputation costs to replace centralized servers. It forces all participants to rationally align with the highest-quality solutions.

Through heterogeneous collaboration in the Agere network, GEB engineering architecture organically interlocks static logical rules with dynamic physical games. By anchoring Bitcoin security, the system establishes sovereignty boundaries; by deterministic engines, it preserves local consistency; and under reputation games, it collapses discrete task solutions into unique linear history. This design enables GEB to transcend the limitations of traditional ledgers and evolve into a network capable of continuous self-organized growth amid uncertainty.

5. GEB Application Example: How Robots See the World Clearly

This chapter uses robotic perception as an example to demonstrate how GEB transforms acquisition of physical signals into an evolutionary process based on **logical collapse**, resolving cognitive conflicts in dynamic environments.

5.1 Cognitive Challenge: Π_2 Propositions in Perception

In dynamic environments, inferring a unique environment model y from noisy signal x is a typical Π_2 proposition. Although the system can verify whether y conforms to physical laws, it cannot predict the one unique reality in advance. Through **sovereign mapping**, GEB defines each sensor as an individual account anchored on Bitcoin UTXO, ensuring the source of perceptual data has economic accountability and non-repudiation.

5.2 Spatial Constraints: "Blockified" Sampling of Environmental Detection

- **Minimum execution unit:** Each independent scan, sample, or displacement of the robot is packaged as a "perception block." Each block contains not only spatial data but also a **deterministic weight** based on node reputation.
- **Turing-machine audit:** The Agere verification engine (VM) performs static audits on blocks (e.g., object displacement must satisfy dynamics). This

establishes a "physical baseline" in the spatial dimension and removes logically absurd claims before games begin.

5.3 Temporal Growth: Divergence Tolerance and Withering Under Ordinal Logic

When perception conflicts arise due to environmental changes (such as opening/closing doors and windows), the system uses ordinal logic to allow **logical forks**.

- **Fork retention:** When new perception conflicts with history, the system does not mechanically overwrite, but retains multiple "candidate history" branches.
- **Dynamic evolution:** Perceptual evidence is recursively superposed onto branches over time. If a branch decouples from physical reality, it stops growing due to lack of subsequent evidence and eventually "withers." This avoids instantaneous cognitive collapse common in traditional AI when facing contradictory data.

5.4 Evolutionary Convergence: Truth Collapse via Reputation Game

This is the key transformation from "multiple possibilities" to "one unique reality":

- **Game convergence:** Different nodes compete to submit perceptual proofs. A validator network composed of high-reputation nodes scores evidence according to completeness and logical consistency.
- **Heaviest-chain collapse:** The path supported by the highest cumulative reputation weight becomes the "heaviest chain." As recursion depth increases, path-conflict probability decays exponentially. As Heisenberg revealed, sustained observation forces the system's uncertain superposition state to collapse into a unique truth highly isomorphic to the physical world.

Robot perception driven by GEB proves that: **truth is not "computed"; it is "evolved" through organizational rules and games with the environment.**

This non-mechanical evolutionary mechanism gives the system endogenous error correction and group coordination capabilities. The robot is no longer an isolated obstacle-avoidance machine, but an intelligent agent that continuously approaches physical truth through adaptive adjustment under rigorous game rules. GEB

realizes spontaneous convergence of complex systems toward truth under concise spacetime-orthogonal rules.

6. Conclusion: The Road to Computable Systems of Group Organization

To break through the computability category crisis widely faced by current AI and blockchain systems due to their roots in a "single formal system," this white paper proposes a brand-new theoretical and engineering framework: GEB.

GEB no longer attempts to forcibly simulate reality using algorithms that pursue absolute **consistency**; instead, it constructs an evolutionary system that is intrinsically **complete**. By integrating the intellectual trajectory from Godel and Turing to Feferman and Nash, it builds a spacetime-orthogonal model for addressing Π_2-level challenges. Through the GEB four-element architecture, this model organically interlocks horizontal spatial games with vertical temporal recursion, laying a theoretical foundation for spontaneous convergence and continuous evolution along ordinal chains in decentralized environments.

In engineering practice, GEB architecture is realized through the heterogeneous collaborative **Agere network**. It natively binds to Bitcoin's sovereignty boundary (individual accounts), solving structural centralization issues at the source. On this basis, each Agere subsystem is a full instantiation of the GEB four elements: **individual accounts** guarantee decentralized sovereign independence; **Turing machines** provide deterministic constraints of physical laws; **ordinal logic** introduces task-driven temporal recursive structures; **non-cooperative games** (such as the reputation convergence operator of Agere0) provide the core collapse driving force, forcing the system to solidify instantaneous probabilistic fluctuations into deterministic historical facts.

This complete collaborative architecture of "verification-computation-time-game" ultimately resolves the logical paradox under a single paradigm: the system no longer depends on a "super-observer" to maintain order, but achieves spontaneous convergence through endogenous conflict-resolution mechanisms. Through the evolutionary example of robotic perception in dynamic environments, we demonstrate this architecture's superior adaptability in handling complex real-world tasks.

GEB theory not only provides a unified perspective for understanding Bitcoin's essence; more importantly, it opens a new path for building the next generation of

adaptive computable systems of group organization. This system will be a collaborative network of decentralized Agere nodes executing various real-world tasks, capable of continuous ascent from logical possibility to physical reality through ongoing games and evolution in open environments.

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